TigerGraph HA Node-Failure - Test Plan

ID	Type	Description	Precondition	Input/Test Steps	Base URL & API
TC-QL-001	Positive	Point lookup: query a Person by primary id returns exactly one vertex.	HA cluster active; sample graph loaded.	1. Open gsql on graph social 2. Run point-lookup query for id p1 3. Verify one vertex returned	gsql -g social 'INTERPRET QUERY(){ ... WHERE v.id=="p1" }'
TC-QL-002	Positive	1-hop traversal: friends of a Person are returned.	HA cluster active; sample graph loaded.	1. Open gsql on graph social 2. Run 1-hop Friendship traversal from p1 3. Verify neighbour set	gsql -g social 'INTERPRET QUERY(){ ... -(Friendship)- Person }'
TC-QL-003	Positive	Aggregation: count of all Person vertices equals 5000.	HA cluster active; sample graph loaded.	1. Open gsql on graph social 2. Run SumAccum count over all Person 3. Verify count == 5000	gsql -g social 'INTERPRET QUERY(){ ACCUM @@n+=1 }'
TC-QL-004	Positive	Multi-hop traversal: 2-hop friendship expansion returns a result set.	HA cluster active; sample graph loaded.	1. Open gsql on graph social 2. Run 2-hop traversal from p1 3. Verify non-empty result	gsql -g social 'INTERPRET QUERY(){ 2-hop }'
TC-SV-001	Positive	Service health: all critical services (GPE, GSE, RESTPP, GSQL) are Online.	HA cluster active.	1. Run gadmin status -v 2. Parse service states 3. Verify all critical services Online	gadmin status -v
TC-LJ-001	Positive	Loading job: load Person rows from a CSV file source; vertex count increases.	HA cluster active; loading job defined.	1. Generate CSV of 200 rows 2. RUN LOADING JOB load_social with the CSV 3. Verify LOAD SUCCESSFUL and count += 200	gsql -g social 'RUN LOADING JOB load_social USING f_person="...csv"'
TC-NF-001	Failure	Data-node hard crash: kill a node; measure availability and MTTR.	HA cluster active; probe running.	1. Start availability probe 2. docker kill tg2 3. Observe availability 4. Recover and measure MTTR	docker kill tg2 ; GET http://localhost:14240/restpp/query/social/ping_count
TC-NF-002	Failure	Graceful node shutdown: stop a node; measure recovery.	HA cluster active; probe running.	1. Start probe 2. docker stop tg2 3. Observe 4. Recover	docker stop tg2 ; GET http://localhost:14240/restpp/query/social/ping_count
TC-NF-003	Failure	Frozen node: pause a node (up but unresponsive); measure recovery.	HA cluster active; probe running.	1. Start probe 2. docker pause tg3 3. Observe 4. Unpause	docker pause tg3 ; GET http://localhost:14240/restpp/query/social/ping_count
TC-NF-004	Failure	Network partition: isolate a node; measure rejoin behaviour.	HA cluster active; probe running.	1. Start probe 2. Disconnect tg3 from network 3. Observe 4. Reconnect	docker network disconnect tgnet tg3 ; GET http://localhost:14240/restpp/query/social/ping_count
TC-NF-005	Failure	Single-component failure: stop GPE on one node; measure behaviour.	HA cluster active; probe running.	1. Start probe 2. gadmin stop GPE_1#2 3. Observe 4. Restart services	gadmin stop GPE ; GET http://localhost:14240/restpp/query/social/ping_count
TC-NF-006	Failure	Master/gateway-node crash: kill tg1; observe from a surviving node.	HA cluster active; probe on tg2.	1. Start probe on tg2 gateway 2. docker kill tg1 3. Observe 4. Recover	docker kill tg1 ; http://localhost:14241/...
TC-SV-002	Failure	Service crash on node loss: killing a node takes its GPE/GSE/RESTPP instances down; a replica stays Online.	HA cluster active.	1. Record baseline gadmin status 2. docker kill tg2 3. Run gadmin status -v 4. Verify tg2 services down and a GPE replica Online	docker kill tg2 ; gadmin status -v

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TC-WR-001	Failure	Write durability under crash: writes during a node crash are not lost.	HA cluster active.	1. Record count 2. docker kill tg3 3. Upsert N vertices via a surviving node 4. Recover; verify no acknowledged write lost	docker kill tg3 ; POST http://localhost:14240/restpp/graph/social
TC-WR-002	Failure	Write durability under partition: ambiguous writes; nothing acknowledged is lost.	HA cluster active.	1. Record count 2. Partition tg2 3. Upsert N vertices 4. Heal; verify durability	network disconnect tg2 ; POST http://localhost:14240/restpp/graph/social
TC-NG-001	Negative	Invalid query is rejected gracefully and does not crash the cluster.	HA cluster active.	1. Run an invalid GSQL query 2. Verify an error is returned 3. Verify the gateway still serves	gsql -g social '' ; GET http://localhost:14240/restpp/query/social/ping_count
TC-BD-001	Boundary	Two-node failure exceeds RF=2 tolerance; cluster must recover after nodes return.	HA cluster active.	1. docker kill tg2 and tg3 2. Observe availability (likely lost) 3. Recover both nodes 4. Verify cluster recovers	docker kill tg2 tg3 ; GET http://localhost:14240/restpp/query/social/ping_count